Dalal Street Pulse

INDIAN STOCK MARKET INTELLIGENCE PLATFORM — ARCHITECTURE & FUTURE SCOPE

Document: Technical & Strategic Whitepaper
Markets: NSE & BSE India (Equity & SME)
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1. Executive Summary & Vision

Dalal Street Pulse is a specialized, institutional-grade financial intelligence and analytics web application engineered specifically for the **Indian equity ecosystem (NSE & BSE)**. Mainstream financial platforms often treat Indian markets as an afterthought, relying on synthetic US feeds, foreign currency units, or inaccurate session rules. Dalal Street Pulse addresses this by delivering native Rupee (`₹`) formatting with authentic Lakh/Crore notation, precise `Asia/Kolkata` (IST) trading session tracking, multi-tier data ingestion reconciling Dual Listings via **ISIN**, and grounded AI-driven equity valuations.

The Problem Solved — INDUSTRY PAIN POINT

- Generic tools lack Lakh/Crore grouping and IST trading session clocks (Pre-open, Muhurat, Normal, Post-closing).
- Disconnected ticker systems that fail to reconcile dual-listed stocks (NSE vs BSE Scrip codes).
- Superficial AI outputs disconnected from SEBI/exchange regulatory disclosures and fundamentals.

The Solution Provided — VALUE DELIVERED

- Zero synthetic data: verified NSE/BSE security masters (Mainboard & SME).
- Grounded AI: fundamental health scoring & peer valuation multiple benchmarking.
- Automated background workflows for price alerts and daily post-market briefings.

2. Production Technology Stack

Layer	Technology	Architectural Role
Frontend Framework	Next.js 16 (App Router + Turbopack)	High-performance server-rendered UI, dynamic route streaming, instant page loads.
Language & Types	TypeScript (Strict Mode)	End-to-end type safety across quotes, OHLCV candles, and valuation metrics.
Styling & UI	Tailwind CSS v4 + shadcn/ui (Nova)	Curated dark-mode financial tokens, responsive layout, Geist typography.
Data Ingestion	Pluggable Hybrid Ingestion Engine	ISIN-based canonical instrument master reconciling NSE tickers & BSE codes.
Persistence	MongoDB + Resilient Memory Fallback	High-throughput connection pooling with zero-dependency offline resilience.
Background Engine	Inngest SDK (Workflows & Crons)	Automated price alert polling, 16:30 IST market briefings, master syncs.
AI & Notifications	Gemini API + Nodemailer SMTP	Grounded valuation analysis, post-market summaries, instant email alerts.

Dalal Street Pulse — Application Modules

CORE PRODUCTION CAPABILITIES & ARCHITECTURE

Platform: Next.js 16 + Inngest + AI
Timezone: Asia/Kolkata (IST)

3. Current Application Modules & Capabilities

Market Radar & Indices `/` `/markets`

- **Live Benchmarks:** NIFTY 50, SENSEX, NIFTY Bank, NIFTY IT ticker tape.
- **IST Session Clock:** Real-time status with countdowns to pre-open (09:00), regular (09:15), and close (15:30).
- **Institutional Activity:** Daily FII & DII cash market net purchases/sales.
- **Sector Heatmaps:** Visual intraday gainers, losers, and 52W breakouts.

Stock Analytics & Terminal `/stocks` `/stocks/[symbol]`

- **Interactive Charts:** Candlestick OHLC & Area charts with volume bars, crosshairs, and timeframes (1D to 5Y).
- **Valuation Ratios:** P/E vs Industry P/E, P/B, ROE, ROCE, D/E, 52W High/Low.
- **Financial Statements:** Quarterly/annual P&L, Balance Sheet, Cash Flow.
- **Ownership Breakdown:** Shareholding patterns (Promoter, FII, DII, Public).

AI Intelligence & Daily Briefs `/insights` `/briefing`

- **Grounded Valuation Verdict:** Automated *Undervalued*, *Fairly Valued*, or *Overvalued* determinations.
- **Fundamental Health Score:** 1–10 composite health score based on leverage, ROCE, and institutional holdings.
- **Automated 16:30 IST Brief:** Inngest-driven post-market daily digest.

Watchlists & Price Alerts `/watchlists` `/alerts`

- **Multi-Watchlist Management:** User-isolated lists with live price sync.
- **Threshold Triggers:** Target price crossings and % swing alerts.
- **Automated Evaluation:** Background cron evaluation with Nodemailer SMTP email card delivery.

News & Macro Intelligence `/news` `/admin`

- **Curated News Wire:** RBI policies, corporate earnings, SEBI regulations.
- **Sentiment Classification:** Bullish, Bearish, and Neutral sentiment badges.
- **System Admin Health:** Real-time MongoDB and provider latency monitor.

Data Reconciler (ISIN Master) DUAL-EXCHANGE

- **NSE & BSE Linkage:** Unified quotes using canonical ISIN identification.
- **SME Support:** Dedicated handling for NSE EMERGE and BSE SME counters.
- **Symbol Alias Engine:** Automatic tracking of historical scrip renames.

Dalal Street Pulse — Future Scope

STRATEGIC EVOLUTION, FEATURE EXPANSION & LONG-TERM PRODUCT ROADMAP

Strategic Horizon: 2026 – 2028
Target: Retail & Pro Institutional Traders
Status: Roadmap Specification

4. Detailed Future Scope & Product Roadmap

The roadmap below outlines the systematic transformation of Dalal Street Pulse from an intelligence portal into an all-in-one terminal for execution, derivatives analysis, algorithmic tooling, and primary market participation.

PHASE 1 1-Click Broker Trading & Execution Gateway

Bridge the gap between market research and live execution by integrating directly with India's top retail stockbrokers via secure OAuth protocols.

- **Licensed Broker Connectors:** Direct order routing via **Zerodha Kite Connect**, **Groww**, **Upstox API**, **Angel One SmartAPI**, and **Dhan HQ**.
- **1-Click Order Execution:** Place CNC (Cash & Carry), MIS (Intraday), and GTT (Good Till Triggered) orders directly from the stock details view.
- **Paper Trading Simulator:** Virtual ₹10,00,000 INR account to allow novice investors to test algorithmic and discretionary strategies in real time without capital risk.
- **Basket & Thematic Orders:** Execute multi-stock thematic baskets (e.g., "Top Nifty IT Leaders" or "High ROCE Green Energy") in a single confirmation.

PHASE 2 NSE Derivatives (F&O) & Options Analytics Terminal

Deliver professional-grade tooling for India's high-volume index and equity options ecosystem.

- **Interactive NSE Option Chain:** Live Open Interest (OI), Change in OI, Put-Call Ratio (PCR), and Max Pain visualization for NIFTY, BANKNIFTY, and equity F&O.
- **Real-Time Option Greeks:** Dynamic computation of Delta, Gamma, Theta, and Vega with Implied Volatility (IV) and IV Percentile (IVP) curves.
- **Options Strategy Builder:** Visual payoff diagrams for multi-leg option strategies (Bull Call Spreads, Straddles, Iron Condors, Calendar Spreads).
- **Futures Build-up Heatmaps:** Automated classification of Long Build-up, Short Covering, Short Build-up, and Long Unwinding across all contracts.

PHASE 3 Autonomous Agentic AI & Portfolio Auditor

Empower investors with personalized, autonomous agentic intelligence that monitors and optimizes their capital.

- **Consolidated Account Statement (CAS) Auditor:** Drag-and-drop NSDL/CDSL CAS statement parsing to automatically audit portfolio risk, sector overweighting, and asset overlap.
- **Automated Tax-Loss Harvesting:** Smart alerts before March 31 financial year-end identifying unrealized short-term and long-term capital losses to offset tax liabilities.
- **Concall & Transcripts Intelligence:** AI audio/text ingestion of quarterly corporate conference calls to synthesize guidance, margin commentary, and management tone in 60-second summaries.
- **Multilingual Voice & Chatbot:** Native support in regional languages (Hindi, Gujarati, Tamil, Marathi, Bengali, Telugu) for widespread financial literacy.

PHASE 4

IPO Central & Primary Market Intelligence

End-to-end tracking of Mainboard and SME Initial Public Offerings.

- **Comprehensive IPO Radar:** Track upcoming, open, and closed IPOs with price bands, issue sizes, lot quantities, and anchor investor allocations.
- **Grey Market Premium (GMP) Aggregator:** Real-time estimated listing gains tracking.
- **Live Subscription Telemetry:** Hourly bidding rates across QIB, NII (HNI), and Retail investor quotas.
- **AI DRHP Summarizer:** Instant breakdown of Draft Red Herring Prospectus risk factors, promoter backgrounds, and proceeds utilization.

PHASE 5

Omnichannel Delivery, Native Mobile & Institutional API

Cross-platform ecosystem expansion and B2B data monetization.

- **Native Mobile Applications:** iOS & Android apps (React Native) with biometric authentication and lock-screen live activity tickers.
- **Omnichannel Alerts:** Instant notifications delivered straight to **WhatsApp** and **Telegram** for target prices, corporate actions, and dividend dates.
- **Institutional Developer API:** WebSocket L1/L2 data access, historical tick query endpoints, and white-label analytical widgets for fintech developers.

5. Strategic Implementation Roadmap Timeline

Timeline	Milestone Focus	Key Deliverables
Q1 – Q2	Execution & Paper Trading	Zerodha Kite & Upstox OAuth integration, Paper Trading Engine, Custom Multi-metric Screener.
Q3	Derivatives & Options Hub	NSE Option Chain, Strategy Payoff Visualizer, Real-Time Greeks & PCR Tracker.
Q4	AI Auditor & IPO Suite	CAS Portfolio Health Auditor, Tax-Loss Harvester, IPO GMP & DRHP Summarizer.
Year 2+	Mobile Apps & Enterprise API	React Native iOS/Android apps, WhatsApp Bot Alerts, Institutional Developer API tier.